



LAB 5 : STEP AND IMPULSE RESPONSE OF LTI SYSTEM USING MATLAB

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Electronics ELYM316

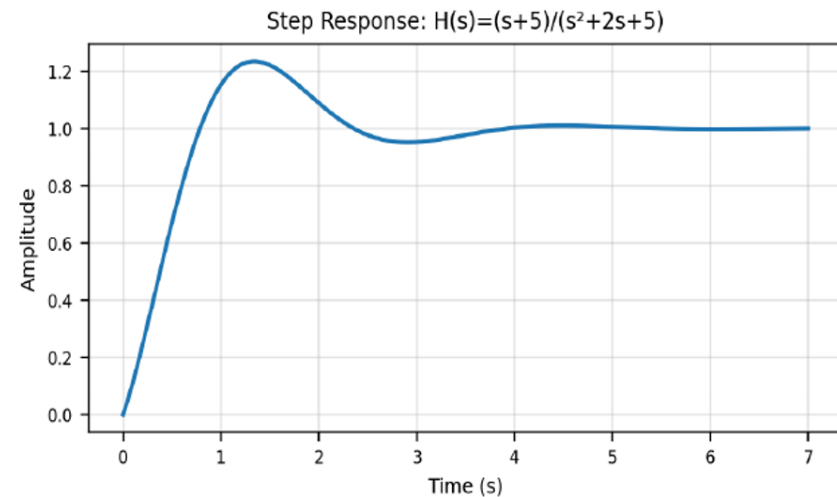
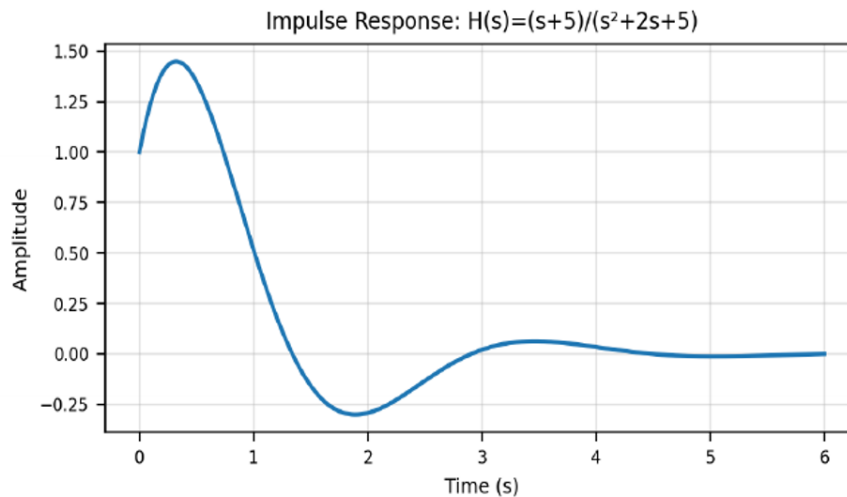
Task 1: Impulse and Step Response

```
% Task 1: Impulse and Step Response
num = [1, 5];          % numerator: s + 5
den = [1, 2, 5];       % denominator: s^2 + 2s + 5
sys = tf(num, den);

figure;
subplot(2,1,1)
impz(sys);
title('Impulse Response of H(s) = (s+5)/(s^2+2s+5)');
grid on;

subplot(2,1,2)
step(sys);
title('Step Response of H(s) = (s+5)/(s^2+2s+5)');
grid on;
```

Observation: The impulse response shows damped oscillatory behaviour due to the complex poles of the transfer function. The step response settles to a steady-state value, confirming that the system is stable.



Task 2: Output Response for Delayed Unit Step Input

```
% Task 2: Output
num2 = [1];
den2 = [1, 3];      % H(s) = 1/(s+3)
sys2 = tf(num2, den2);

t = 0:0.01:10;      % time vector

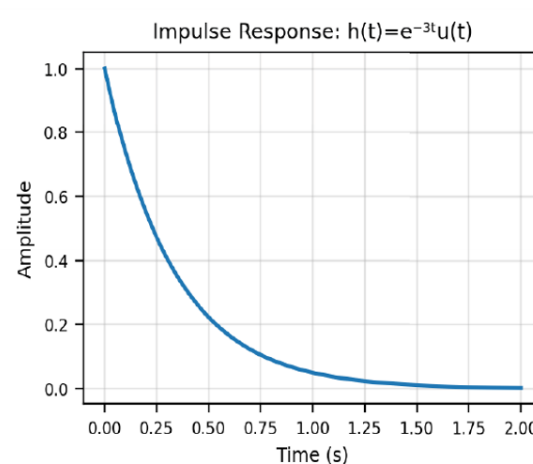
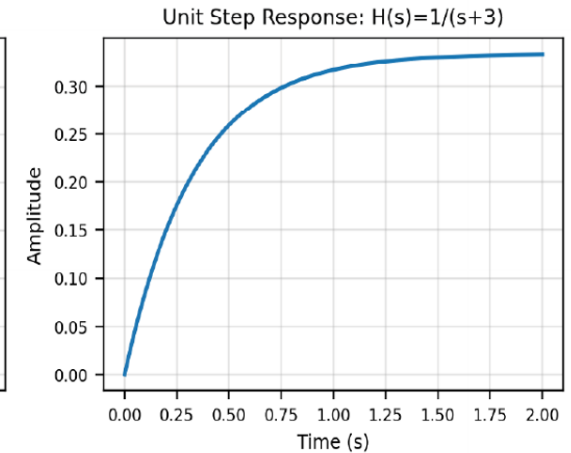
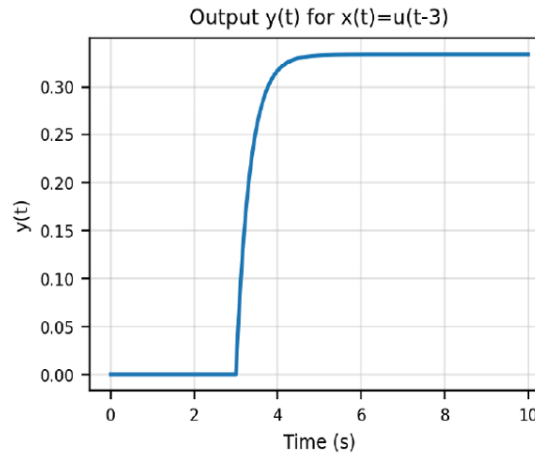
% x(t) = u(t-3): zero for t < 3, one for t >= 3
x = (t >= 3);

% Compute output using lsim
y = lsim(sys2, x, t);

% --- Figure 1: Output y(t) ---
figure;
plot(t, y, 'b', 'LineWidth', 1.5);
title('Output y(t) for x(t) = u(t-3)');
xlabel('t (s)');
ylabel('y(t)');
grid on;

% --- Figure 2: Unit Step Response ---
figure;
step(sys2);
title('Unit Step Response of H(s) = 1/(s+3)');
grid on;

% --- Figure 3: Impulse Response ---
figure;
impz(sys2);
title('Impulse Response h(t) = e^{-3t}u(t)');
grid on;
```



Observation: The output remains zero until $t = 3$ because the delayed unit step input activates at that instant. The response then rises exponentially toward the steady-state value of $1/3$.

Conclusion

MATLAB was successfully used to simulate and analyse the step and impulse responses of LTI systems. The obtained responses matched the expected theoretical behaviour of stable systems.